

Problem 3

How many positive perfect squares less than 2023 are divisible by 5?

- (A) 8 (B) 9 (C) 10 (D) 11 (E) 12

Problem 6

When the roots of the polynomial

$$P(x) = (x - 1)^1(x - 2)^2(x - 3)^3 \cdots (x - 10)^{10}$$

are removed from the number line, what remains is the union of 11 disjoint open intervals. On how many of these intervals is $P(x)$ positive?

- (A) 3 (B) 7 (C) 6 (D) 4 (E) 5

Problem 10

In the xy -plane, a circle of radius 4 with center on the positive x -axis is tangent to the y -axis at the origin, and a circle with radius 10 with center on the positive y -axis is tangent to the x -axis at the origin. What is the slope of the line passing through the two points at which these circles intersect?

- (A) $\frac{2}{7}$ (B) $\frac{3}{7}$ (C) $\frac{2}{\sqrt{29}}$ (D) $\frac{1}{\sqrt{29}}$ (E) $\frac{2}{5}$

Problem 12

For complex numbers $u = a + bi$ and $v = c + di$, define the binary operation $\otimes$ by

$$u \otimes v = ac + bdi.$$

Suppose z is a complex number such that $z \otimes z = z^2 + 40$. What is $|z|$?

- (A) 2 (B) 5 (C) $\sqrt{5}$ (D) $\sqrt{10}$ (E) $5\sqrt{2}$